SVKM'S NMIMS

MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT ENGINEERING / SCHOOL OF TECHNOLOGY MANAGEMENT ENGINEERING

Program: B.Tech / MBA Tech (All Stream) Year: III Semester : V

AcademicYear: 2022-2023

Subject: Renewable Energy Sources

Date : 07 February 2023

Marks: 100

Time: 2.00 pm - 5.00 pm

Durations: 3 (Hrs)

No. of Pages: 2

Re-Examination (2022-23)

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

1) Question No. 1 is compulsory.
2) Out of remaining questions, attempt any 4 questions.
3) **In all 5 questions to be attempted.**
4) All questions carry equal marks.
5) **Answer to each new question to be started on a fresh page.**
6) **Figures in brackets on the right hand side indicate full marks.**
7) **Assume Suitable data if necessary.**

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Q1		Answer briefly:	[20]
CO- 1; SO-7 ; BL-2	a.	Discuss different renewable and non-renewable energy sources.	
CO- 2; SO-7 ; BL-3	b.	Elaborate benefits and application of OTEC.	
CO-3 ; SO- 7; BL-4	c.	Explain initial and annual cost.	
CO-2 ; SO-7 ; BL-2	d.	Discuss main advantages and disadvantages of biomass energy.	
Q2 CO-2; SO-7; BL-4	a.	Illustrate different Solar angles with neat sketch.	[10]
CO-3; SO-7; BL-3	b.	Elaborate Cash flow diagram with neat sketch.	[10]
Q3 CO-2; SO-7; BL-4	a.	Illustrate working of Biomass-fired steam power plant.	[10]
CO-3; SO-1; BL-4	b.	Determine the sunset hour angle and day length at a location latitude of 32^0 on March 31. The standard longitude for IST is 81^0 and 44' E.	[10]

Q4 CO-2; SO-7; BL-4	a.	Illustrate construction and working of Hydro-electric power plant with neat sketch.	[10]
CO-2; SO-7; BL-4	b.	Explain Dry steam power plant with neat sketch.	[10]
Q5 CO-3; SO-1; BL-4	a.	A Horizontal axis wind turbine has following data: Speed of free wind at a height of 10 m = 13 m/s Air density = 1.226 kg/m^3 $\alpha = 0.15$ Height of tower = 100 m Diameter of rotor = 80 m Wind velocity at the turbine reduces by 20% Generator efficiency = 85% Find: (i) Total power available in wind (ii) Power extracted by the turbine (iii) Electrical power generated	[10]
CO- 1; SO-7 ; BL-2	b.	Discuss different energy alternatives to fossil fuels.	[10]
Q6 CO-2; SO-7; BL-4	a.	Illustrate working of Solar dryer with neat sketch.	[10]
CO-2; SO-7; BL-4	b.	Explain different types of bio-fuels.	[10]
Q7 CO-2; SO-7; BL-3	a.	Elaborate components and working of Horizontal axis wind turbine with neat sketch.	[10]
CO-2; SO-7; BL-4	b.	Explain Pitching Type wave energy conversion devices.	[10]

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MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT& ENGINEERING/

SCHOOL OF TECHNOLOGY MANAGEMENT

Academic Year: 2024-2025

Program/s: B TECH / MBA TECH/ BTI

Year: III/V Semester: V/IX

Stream/s ; All applicable streams

Subject: Renewable Energy Sources

Time:03 hrs (2:00pm to 5:00pm)

Date: 10 / 12 / 2024

No. of Pages: 03

Marks: 100

Final Examination/Re-Examination (2023-24) (2024-25)

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

1) Question No. 1 is compulsory.
2) Out of remaining questions, attempt any 4 questions.
3) **In all 5 questions to be attempted.**
4) All questions carry equal marks.
5) **Answer to each new question to be started on a fresh page.**
6) **Figures in brackets on the right-hand side indicate full marks.**
7) **Assume Suitable data if necessary.**

Q1		Answer briefly:	[20]
CO- 1; SO-7 ; BL-2	a.	Discuss Energy classification in details.	
CO- 2; SO-7 ; BL-3	b.	Elaborate Hot dry rock system with neat sketch.	
CO-3 ; SO- 7; BL-2	c.	Explain different types of costs.	
CO-2 ; SO-7 ; BL-2	d.	Explain different types of bio-fuels.	
Q2 CO-2; SO-1; BL-4	a.	Calculate the angle made by beam radiation with the normal to a flat plate collector on March 15, at 10 A.M. solar time for a location at 35° 45' N. The collector is tilted at an angle of 45° with the horizontal and is pointing due south.	[10]

<table>
<tr>
<td>CO-3; SO-1;
BL-3</td>
<td>b.</td>
<td>Evaluate the financial viability of two proposed solar power projects using the NPV method. Both projects have the same capital cost but different cash flows over time. The discount rate is 6% for both projects.
<table>
<tr><th></th><th>Project A</th><th>Project B</th></tr>
<tr><td>Capital cost (Rs.)</td><td>45,000</td><td>45,000</td></tr>
<tr><td>Year</td><td>Net annual saving (Rs.)</td><td>Net annual saving (Rs.)</td></tr>
<tr><td>1</td><td>10,000</td><td>9,500</td></tr>
<tr><td>2</td><td>9,500</td><td>9,300</td></tr>
<tr><td>3</td><td>9,800</td><td>9,000</td></tr>
<tr><td>4</td><td>9,500</td><td>9,200</td></tr>
<tr><td>5</td><td>9,000</td><td>9,000</td></tr>
</table>
i. Calculate the NPV for both projects.
ii. Determine which project is more financially viable based on the NPV values.</td>
<td>[10]</td>
</tr>
<tr>
<td>Q3
CO-2; SO-7;
BL-4</td>
<td>a.</td>
<td>Elaborate working of Fixed dome biogas plant with neat sketch. Also, discuss the advantages and disadvantages of biomass energy.</td>
<td>[10]</td>
</tr>
<tr>
<td>CO-3; SO-7;
BL-4</td>
<td>b.</td>
<td>Illustrate and explain any five solar angles with a neat sketch.</td>
<td>[10]</td>
</tr>
<tr>
<td>Q4
CO-2; SO-7;
BL-4</td>
<td>a.</td>
<td>Classify different hydraulic turbines. Also, elaborate working of Vertical axis wind turbine with neat sketch.</td>
<td>[10]</td>
</tr>
<tr>
<td>CO-2; SO-7;
BL-4</td>
<td>b.</td>
<td>Elaborate Heaving and Pitching Float Type wave energy devices with neat sketch. Also state applications of wave energy.</td>
<td>[10]</td>
</tr>
</table>

Q5 CO-3; SO-1; BL-4	a.	The following data were collected for a three-blade Horizontal Axis Wind Turbine (HAWT): Average wind speed at a standard height of 10 m = 12 m/s Wind shear coefficient α=0.14 Air density = 1.225 kg/m^3 Hub height = 90 m Rotor diameter = 70 m The downstream wind velocity is 60% of the upstream wind velocity. Estimate the power available in the wind, power extracted by the wind turbine, and axial force acting on the wind turbine.	[10]
CO- 1; SO-7 ; BL-2	b.	Discuss Nuclear and Solar energy alternatives to fossil fuels with advantages and disadvantages.	[10]
Q6 CO-2; SO-7; BL-4	a.	Illustrate the working of the Solar water heating system with neat sketch.	[10]
CO-2; SO-7; BL-4	b.	Illustrate and explain the working of the Biomass Cogeneration power plant with a neat sketch.	[10]
Q7 CO-2; SO-7; BL-3	a.	Elaborate basic components of the wind energy conversion system with a neat sketch.	[10]
CO-2; SO-7; BL-4	b.	Explain the working of the Open cycle OTEC plant. List down the limitations of Ocean energy.	[10]

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MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT& ENGINEERING/

SCHOOL OF TECHNOLOGY MANAGEMENT

Academic Year: 2023-2024

Program/s: B Tech/MBA Tech/B Tech Intg

Year: III/V Semester: V/IX

Stream/s ;All Applicable Streams

Subject: Renewable Energy Sources

Time: 03 hrs (2 pm to 5 pm)

Date: 16/02/2024

No. of Pages: 03

Marks: 100

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ReExamination

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

1) Question No. 1is compulsory.
2) Out of remaining questions, attempt any 4 questions.
3) **In all 5 questions to be attempted.**
4) All questions carry equal marks.
5) **Answer to each new question to be started on a fresh page.**
6) **Figures in brackets on the right hand side indicate full marks.**
7) **Assume Suitable data if necessary.**

Q1		Answer briefly:	[20]
CO- 1;SO-7; BL-4	a.	Elaborate Global energy crises.	
CO-3;SO-7 ; BL-3	b.	Interpret working of Solar dryer with neat sketch.	
CO-1; SO- 7;BL-1	c.	Describe different forms of Biomass energy.	
CO-3 ;SO-7 ; BL-4	d.	Explain Return on Investment (ROI) with advantages and disadvantages.	
Q2 CO-2; SO-7; BL-3	a.	Interpret different Biomass characteristics in detail.	[7]
CO-3; SO-7; BL-3	b.	Elaborate working of Kaplan turbine with neat sketch.	[7]
CO-1; SO-7; BL-4	c.	Explain different Renewable energy sources with advantages and disadvantages	[6]
Q3 CO-3; SO-1; BL-5	a.	Estimate Sunset hour angle, and daylight hours (sunshine hours) at Kanpur (latitude 26.45° North) on June 28th in a leap year.	[8]

CO-3; SO-7; BL-4	b.	Explain working of Oscillating water column wave energy conversion device with neat sketch.	[7]
CO-3; SO- 7; BL-4	c.	Explain Payback Period with advantages and disadvantages.	[5]
Q4 CO-3;SO-1; BL-4	a.	A wind turbine (VAWT) is to be installed at a tower of height 120 m in a locality. Considering the given data for the system Calculate i. Speed of wind at a tower height ii. Total power available in wind iii. Interference factor iv. Power coefficient v. Power extracted by the turbine vi. Electrical power generated by the generator Given: 1. Speed of the free wind at a height of 10 m = 15 m/s 2. Air density = 1.226 kg/m^3 3. α (constant) = 0.14 4. Diameter of rotor = 100 m 5. wind velocity at the turbine reduces by 25% 6. Generator efficiency = 90%	[8]
CO-3;SO-1; BL-4	b.	A family of 4 adults plan to install a biogas plant for their cooking and lighting needs with seven 100 CP lamps for six hours daily. Calculate: (i) Number of cows required to feed the biogas plant (ii) Volume of the digester Consider the data given below: • Cow dung production rate is 15 kg /day/head with Gas yield of 0.40 m^3 per kg of dry matter. • Only 60% of the cow dung produced is collectible. • Solid (dry) matter content in cow dung is about 18% only. • Biogas required for cooking is about 0.35 m^3/person/day • Gas required for lighting of 100 CP (candle Power) mantle lamp is 0.15 m^3/hour. • The density of slurry is about 1090 kg/m^3. • Consider 40-day retention time for volume of slurry in the digester and 90% occupied. • Heating value of biogas = 23MJ/m^3	[8]
CO-1;SO-7; BL-2	c.	Discuss need of renewable energy sources in India.	[4]
Q5 CO-3;SO-7; BL-4	a.	Elaborate working of Dry steam and Wet Steam Geothermal plant with neat sketch.	[10]

<table>
<tr>
<td>CO-3;
SO-1;
-5</td>
<td>b.</td>
<td>Two proposed projects are presented in the table given below:
<table>
<tr><th></th><th><u>Project 1</u></th><th><u>Project 2</u></th></tr>
<tr><td>Capital cost (Rs.)</td><td>50,000</td><td>50,000</td></tr>
<tr><td>Year</td><td>Net annual saving (Rs.)</td><td>Net annual saving (Rs.)</td></tr>
<tr><td>1</td><td>+8,000</td><td>+8,600</td></tr>
<tr><td>2</td><td>+8,000</td><td>+8,600</td></tr>
<tr><td>3</td><td>+8,000</td><td>+8,300</td></tr>
<tr><td>4</td><td>+8,000</td><td>+8,300</td></tr>
<tr><td>5</td><td>+8,000</td><td>+8,000</td></tr>
<tr><td>6</td><td>+8,000</td><td>+8,000</td></tr>
<tr><td>7</td><td>+8,000</td><td>+7,700</td></tr>
<tr><td>8</td><td>+8,000</td><td>+7,700</td></tr>
<tr><td>9</td><td>+8,000</td><td>+7,400</td></tr>
<tr><td>10</td><td>+8,000</td><td>+7,400</td></tr>
<tr><td>Total net savings at end of 10th year</td><td>80,000</td><td>80,000</td></tr>
</table>
Evaluate the financial viability of two proposed projects by using the Net Present Value (NPV) analysis technique. Both projects are subject to an annual discount rate of 6%.</td>
<td>[10]</td>
</tr>
<tr>
<td>Q6
CO-1;SO-7;BL-4</td>
<td>a.</td>
<td>Classify different solar collectors and explain working of Flat plate solar collector with neat diagram.</td>
<td>[7]</td>
</tr>
<tr>
<td>CO-3;SO-7;BL-4</td>
<td>b.</td>
<td>Elaborate working of Open Cycle Ocean Thermal Energy Conversion system with neat sketch.</td>
<td>[7]</td>
</tr>
<tr>
<td>CO-1;SO-7;BL-4</td>
<td>c.</td>
<td>Explain advantages, disadvantages and applications of Wind Energy.</td>
<td>[6]</td>
</tr>
<tr>
<td>Q7
CO-1;SO-7;BL-4</td>
<td>a.</td>
<td>Elaborate different angles made by solar radiations with surface of the earth.</td>
<td>[10]</td>
</tr>
<tr>
<td>CO-1;SO-7;BL-3</td>
<td>b.</td>
<td>Interpret basic components of wind energy conversion system with neat sketch.</td>
<td>[10]</td>
</tr>
</table>

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MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT& ENGINEERING/

SCHOOL OF TECHNOLOGY MANAGEMENT

Academic Year: 2024-2025

Program/s: B TECH / MBA TECH/BTI

Year: III/V Semester: V/IX

Stream/s ; All applicable streams

Subject: Renewable Energy Sources

Time:03 hrs (10:00am to 1:00pm)

Date: 25/2/2025

No. of Pages: 03

Marks: 100

Re-Examination (2023-24)/24-25

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

1) Question No. 1 is compulsory.
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6) **Figures in brackets on the right-hand side indicate full marks.**
7) **Assume Suitable data if necessary.**

Q1		Answer briefly:	[20]
CO- 1; SO-7 ; BL-2	a.	Discuss the present energy status of India and the world.	
CO- 2; SO-7 ; BL-3	b.	Elaborate benefits and application of the Ocean Thermal Energy Conversion system.	
CO-3 ; SO- 7; BL-2	c.	Explain different Financial Analysis Techniques.	
CO-2 ; SO-7 ; BL-2	d.	Explain different types of Biomass characteristics.	
Q2 CO-2; SO-1; BL-4	a.	Determine the angle made by beam radiation with the normal to a solar collector on July 25, at 11 A.M. solar time for a location at 40° 20' N. The solar collector is inclined at an angle of 30° with respect to the horizontal and is facing due south.	[10]

<table>
<tr>
<td>CO-3; SO-1;
BL-3</td>
<td>b.</td>
<td>Evaluate the financial viability of two proposed solar power projects using the NPV method. Both projects have the same capital cost but different cash flows over time. The discount rate is 5% for both projects.
<table>
<tr><td></td><td><u>Project A</u></td><td><u>Project B</u></td></tr>
<tr><td>Capital cost (Rs.)</td><td>40,000</td><td>40,000</td></tr>
<tr><td>Year</td><td>Net annual saving (Rs.)</td><td>Net annual saving (Rs.)</td></tr>
<tr><td>1</td><td>8900</td><td>8600</td></tr>
<tr><td>2</td><td>8000</td><td>8600</td></tr>
<tr><td>3</td><td>8900</td><td>8300</td></tr>
<tr><td>4</td><td>8000</td><td>8300</td></tr>
<tr><td>5</td><td>8000</td><td>8000</td></tr>
<tr><td>6</td><td>8000</td><td>8000</td></tr>
</table>
i. Calculate the NPV for both projects.
ii. Determine which project is more financially viable based on the NPV values.</td>
<td>[10]</td>
</tr>
<tr>
<td>Q3
CO-2; SO-7;
BL-4</td>
<td>a.</td>
<td>Elaborate on the working of the Floating dome biogas plant with a neat sketch. Also, discuss Applications of biomass energy.</td>
<td>[10]</td>
</tr>
<tr>
<td>CO-3; SO-7;
BL-4</td>
<td>b.</td>
<td>Illustrate and explain the Liquid flat plate solar collector with a neat sketch. Also, classify different solar collectors.</td>
<td>[10]</td>
</tr>
<tr>
<td>Q4
CO-2; SO-7;
BL-4</td>
<td>a.</td>
<td>Elaborate working of the Francis turbine with a neat sketch. Also, explain the advantages and disadvantages of Wind energy.</td>
<td>[10]</td>
</tr>
<tr>
<td>CO-2; SO-7;
BL-4</td>
<td>b.</td>
<td>Illustrate Double flash liquid-dominated geothermal system with a neat sketch. Also state applications of Ocean energy.</td>
<td>[10]</td>
</tr>
</table>

Q5 CO-3; SO-1; BL-4	a.	For a four-blade HAWT, the following observations were made: Average wind speed at a standard height of 10 m = 10 m/s Wind shear coefficient α=0.15 Air density = 1.23 kg/m^3 Hub height = 100 m Rotor diameter = 80 m The downstream wind velocity is 70% of the upstream wind velocity i. Compute the power available in the wind. ii. Determine the power extracted by the wind turbine. iii. Calculate the axial force exerted on the wind turbine.	[10]
CO- 1; SO-7 ; BL-2	b.	Discuss Tar sand and Oil shale alternatives to fossil fuels with advantages and disadvantages.	[10]
Q6 CO-2; SO-7; BL-4	a.	Discuss any two applications of Solar energy systems with neat sketch.	[10]
CO-2; SO-7; BL-4	b.	Illustrate and explain the working of the Biomass Combined cycle power plant with a neat sketch.	[10]
Q7 CO-2; SO-7; BL-3	a.	Elaborate basic components and working Horizontal axis wind turbine with a neat sketch.	[10]
CO-2; SO-7; BL-4	b.	Explain the working of the Closed cycle OTEC plant. List down the advantages of Ocean energy.	[10]

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MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT& ENGINEERING/

SCHOOL OF TECHNOLOGY MANAGEMENT

Academic Year: 2023-2024

Program/s: B Tech/MBA Tech/B Tech Intg

Year: III/V Semester: V/IX

Stream/s ;All Applicable Streams

Subject: Renewable Energy Sources

Time: 03 hrs (2 pm to 5 pm)

Date: 04 /12/ 2023

No. of Pages: 03

Marks: 100

Final Examination

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

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5) **Answer to each new question to be started on a fresh page.**
6) **Figures in brackets on the right hand side indicate full marks.**
7) **Assume Suitable data if necessary.**

Q1		Answer briefly:	[20]
CO- 1;SO-7; BL-4	a.	Explain Solar energy as an alternative energy source with advantages and disadvantages.	
CO-3;SO-7 ; BL-3	b.	Discuss working of Solar water heater with neat sketch.	
CO-1; SO- 7;BL-4	c.	Explain the functioning of a downdraft gasifier, accompanied by a detailed sketch.	
CO-3 ;SO-7 ; BL-4	d.	Interpret Net Present Value briefly.	
Q2 CO-3; SO-7; BL-4	a.	Explain the operation of a Combined cycle power plant with power generation capabilities.	[7]
CO-3; SO-7; BL-4	b.	Interpret the design and operation of a hydro-electric power plant using a clear diagram.	[7]
CO-3; SO-7; BL-4	c.	Analyse various renewable and non-renewable energy sources and their distinctions.	[6]

Q3 CO-3; SO-1; BL-5	a.	Estimate the sunset hour angle and the duration of daylight hours (sunshine hours) at Bhubaneswar, which is located at a latitude of 20.29° North, on April 10th in a non-leap year.	[8]
CO-3; SO-7; BL-4	b.	Elaborate functioning of Pitching-type wave energy conversion device with neat sketch.	[7]
CO-3; SO- 7; BL-4	c.	Discuss initial and annual cost briefly.	[5]
Q4 CO-3;SO-1; BL-4	a.	Following data were measured for a HAWT: Speed of wind = 15 m/s at 1 atm and 27 °C Diameter of rotor = 100 m Speed of rotor = 45 rpm Calculate the torque produced at the shaft for maximum output of the turbine.	[8]
CO-3;SO-1; BL-4	b.	Calculate the volume of a fixed dome type biogas digester for the output of 10 cows. Also calculate the thermal power available from biogas. Use following data. Retention time: 40 days, Dry matter produced: 3kg/day/cow Biogas yield: 0.30 m^3/kg of dry matter, % of dry matter in cow dung:18% Density of slurry:1090 kg/m^3, Burner efficiency:70% Heating value of biogas: 23MJ/m^3 % volume of the digester occupied by slurry: 90%	[8]
CO-1;SO-7; BL-2	c.	Discuss the current energy scenario in the world.	[4]

<table>
<tr>
<td>Q5
CO-3;
SO-1;
BL-5</td>
<td>a.</td>
<td>Conduct a Net Present Value (NPV) analysis for the two proposed projects presented in the table below, considering an annual discount rate of 5%, to assess their financial viability.
<table>
<tr><td></td><td><u>Project 1</u></td><td><u>Project 2</u></td></tr>
<tr><td>Capital cost (Rs.)</td><td>50,000</td><td>50,000</td></tr>
<tr><td>Year</td><td>Net annual saving (Rs.)</td><td>Net annual saving (Rs.)</td></tr>
<tr><td>1</td><td>+12,000</td><td>+15,000</td></tr>
<tr><td>2</td><td>+12,000</td><td>+15,000</td></tr>
<tr><td>3</td><td>+12,000</td><td>+12,000</td></tr>
<tr><td>4</td><td>+12,000</td><td>+12,000</td></tr>
<tr><td>5</td><td>+12,000</td><td>+6,000</td></tr>
<tr><td>Total net savings at end of 10th year</td><td>60,000</td><td>60,000</td></tr>
</table>
</td>
<td>[10]</td>
</tr>
<tr>
<td>CO-3;SO-7; BL-4</td>
<td>b.</td>
<td>Elaborate working of Single Flash and double Flash Geothermal plant with neat sketch.</td>
<td>[10]</td>
</tr>
<tr>
<td>Q6
CO-1;SO-7;BL-4</td>
<td>a.</td>
<td>Illustrate Photovoltaic cell technologies and elucidate their advantages.</td>
<td>[7]</td>
</tr>
<tr>
<td>CO-3;SO-7;BL-4</td>
<td>b.</td>
<td>Elaborate working of Closed Cycle Ocean Thermal Energy Conversion system with neat sketch.</td>
<td>[7]</td>
</tr>
<tr>
<td>CO-3;SO-7;BL-4</td>
<td>c.</td>
<td>Explain on the components and operation of a Vertical axis wind turbine, complemented by an illustrative sketch.</td>
<td>[6]</td>
</tr>
<tr>
<td>Q7
CO-1;SO-7;BL-4</td>
<td>a.</td>
<td>Draw and Explain Solar radiation geometry.</td>
<td>[10]</td>
</tr>
<tr>
<td>CO-1;SO-7;BL-3</td>
<td>b.</td>
<td>Classify Hydraulic turbines and discuss the working of Horizontal axis wind turbine with neat sketch.</td>
<td>[10]</td>
</tr>
</table>

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Programme: B. Tech (AI)/B. Tech (COMP)/B. Tech (CSE - CS)/B. Tech (EXTC)/B. Tech (IT)/B. Tech (MECH)/B. Tech (MXTC)/MBA Tech (AI)/MBA Tech (Comp)/MBA Tech (DS)/MBA Tech (IT)
Year: III/Semester V(Exam Year: 2025-2026)

Subject: Renewable Energy Sources
Date: 10 Dec 2025 **Time: 02:00 pm to 05:00 pm (03:00 Hrs.)**
Max. Marks: 100

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FINAL EXAMINATION (Acad. Year:**2025-2026**)

Instructions:
1. This question paper contains 3 pages.
2. Answer to each new question to be started on a fresh page.
3. Figure in right hand side indicates full marks
4. Figures in brackets on the right-hand side indicate the maximum marks for each question.
5. Assume suitable data wherever necessary.
6. Draw neat and properly labeled diagrams wherever applicable.

Additional Instructions:
1) Question No. 1 is compulsory.
2) Out of remaining questions, attempt any 4 questions.
3) **In all 5 questions to be attempted**

Q1		Answer briefly:	[20]
CO- 1; SO-7 ; BL-2	a.	Discuss different energy alternatives to fossil fuels.	
CO- 2; SO-7 ; BL-3	b.	Elaborate benefits and applications of OTEC.	
CO-3 ; SO- 7; BL-2	c.	Explain initial and annual cost.	
CO-2 ; SO-7 ; BL-2	d.	Discuss different forms of biomass.	
Q2 CO-2; SO-1; BL-4	a.	For New Delhi (28° 35' N, 77° 12' E), calculate the zenith angle of the sun at 2:30 P.M. on 20 February 2015. The standard IST latitude for India is 81° 44' E.	[10]

<table>
<tr><td>CO-3; SO-1;
BL-3</td><td>b.</td><td>Compare projects A and B using the given data. Use the Net Present Value (NPV) method of evaluation.
<table><tr><td></td><td><u>Project A</u></td><td><u>Project B</u></td></tr><tr><td>Capital cost (Rs.)</td><td>10,00,000</td><td>10,00,000</td></tr><tr><td>Cost of capital</td><td>15%</td><td>13%</td></tr><tr><td>Year</td><td>Net annual saving (Rs.)</td><td>Net annual saving (Rs.)</td></tr><tr><td>1</td><td>2,00,000</td><td>3,00,000</td></tr><tr><td>2</td><td>3,00,000</td><td>4,00,000</td></tr><tr><td>3</td><td>4,00,000</td><td>4,00,000</td></tr><tr><td>4</td><td>3,00,000</td><td>3,00,000</td></tr><tr><td>5</td><td>1,00,000</td><td>2,00,000</td></tr></table>i. Calculate the NPV for both projects.
ii. Determine which project is more financially viable based on the NPV values.</td><td>[10]</td></tr>
<tr><td>Q3
CO-2; SO-7;
BL-4</td><td>a.</td><td>Calculate the volume of a cow dung-based biogas plant required for cooking needs of a family of five adults and lighting need with two 100 C.P. lamps for three hours daily. Also calculate the required number of cows to feed the plant. Assume standard values of data where required.
Following data may be considered for BIOMASS Numerical:
• Cow dung production rate is 10 kg /day/head with Gas yield of 0.34 m^3 per kg of dry matter.
• Only 70% of the cow dung produced is collectible.
• Solid matter content in cow dung is about 18% only.
• Biogas required for cooking is about 0.227 m^3/person/day
• Gas required for lighting of 100 CP (candle Power) mantle lamp is 0.126 m^3/hour.
• The density of slurry is about 1090 kg/m^3.
• Consider 50-day retention time for volume of slurry in the digester and 90% occupied.
• Heating value of biogas = 23MJ/m^3</td><td>[10]</td></tr>
<tr><td>CO-3; SO-7;
BL-4</td><td>b.</td><td>Illustrate and explain Solar Geometry with a neat sketch.</td><td>[10]</td></tr>
<tr><td>Q4
CO-2; SO-7;
BL-4</td><td>a.</td><td>Elaborate working of Vertical axis wind turbine with neat sketch. Also write down pros and cons of Wind Energy.</td><td>[10]</td></tr>
<tr><td>CO-2; SO-7;
BL-4</td><td>b.</td><td>Interpret different Geothermal energy resources with neat sketch.</td><td>[10]</td></tr>
</table>

Q5 CO-3; SO-1; BL-4	a.	Illustrate construction and working of Hydro-electric power plant with neat sketch. Also explain global and local wind.	[10]
CO- 1; SO-7 ; BL-2	b.	Discuss different alternatives to fossil fuels with advantages and disadvantages.	[10]
Q6 CO-2; SO-7; BL-4	a.	Illustrate the working of the Liquid Flat plate collector with neat sketch.	[10]
CO-2; SO-7; BL-4	b.	Elaborate working of Floating drums biogas plant with neat sketch. Also state different applications of biomass energy.	[10]
Q7 CO-2; SO-7; BL-3	a.	Classify different Hydraulic turbines. Elaborate working of basic components of Wind Energy Conversion System with neat sketch.	[10]
CO-2; SO-7; BL-4	b.	Analyse working of Closed cycle Ocean Thermal Energy Conversion (OTEC) system with neat sketch. List down the limitations of Ocean energy.	[10]

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MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING / SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING

Academic Year: 2022-23

Programme: B.Tech / MBA Tech (All Stream) Year: III Semester: V

Subject: Renewable Energy Sources

Date: 01 December 2022

Marks: 100
Time: 2.00 pm - 5.00 pm
Durations: 3 (Hrs)
No. of Pages: 02

Final Examination

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

1) Question No. 1 is compulsory.
2) Out of remaining questions, attempt any 4 questions.
3) **In all 5 questions to be attempted.**
4) All questions carry equal marks.
5) **Answer to each new question to be started on a fresh page.**
6) **Figures in brackets on the right hand side indicate full marks.**
7) **Assume Suitable data if necessary.**

Q1		Answer briefly:	[20]
CO- 1; SO-7 ; BL-2	a.	Discuss present energy scenario of India and world.	
CO- 2; SO-7 ; BL-3	b.	Elaborate benefits and application of Geothermal energy.	
CO-3 ; SO- 7; BL-4	c.	Explain types of costs.	
CO-2 ; SO-7 ; BL-2	d.	Discuss different forms of biomass.	
Q2 CO-2; SO-7; BL-4	a.	Illustrate Photovoltaic cell technologies. Also explain its advantages.	[10]
CO-3; SO-7; BL-3	b.	Elaborate Cash flow diagram with neat sketch.	[10]
Q3 CO-2; SO-7; BL-4	a.	Illustrate working of combined cycle power plant power plant.	[10]
CO-3; SO-1; BL-4	b.	Determine the sunset hour angle and day length at a location latitude of 33^0 on March 29. The standard longitude for IST is 81^0 and 44' E.	[10]

Q4 CO-2; SO-7; BL-4	a.	Illustrate construction and working of Pelton wheel turbine with neat sketch.	[10]
CO-2; SO-7; BL-4	b.	Explain Single flash geothermal power plant with neat sketch.	[10]
Q5 CO-3; SO-1; BL-4	a.	A HAWT has following data: Speed of wind =15 m/s at 1 atm and 15 °C Diameter of rotor = 125 m Speed of rotor = 45 rpm Calculate the maximum possible torque produced at the shaft.	[10]
CO- 1; SO-7 ; BL-2	b.	Discuss the need of renewable energy sources. List the demerits of different non-renewable energy sources.	[10]
Q6 CO-2; SO-7; BL-4	a.	Illustrate working of Compound parabolic collector with neat sketch.	[10]
CO-2; SO-7; BL-4	b.	Explain Downdraft gasifier with neat sketch.	[10]
Q7 CO-2; SO-7; BL-3	a.	Elaborate components and working of Vertical axis wind turbine with neat sketch.	[10]
CO-2; SO-7; BL-4	b.	Explain Heaving float Type wave energy conversion devices.	[10]

X